UNIVERSITY OF GHANA

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE
STAT 112: INTRODUCTION TO STATISTICS AND PROBABILITY
EXERCISE II

1. A commuter travels to work each morning and takes exactly one of three possible routes: Route A (motorway), Route B (through town), and Route C (a back-road shortcut). On any given workday, the probability that the commuter selects Route A is 50%, the probability of choosing Route B is 30%, and there is a 20% chance of taking Route C. Traffic records show the following delays: *if* the commuter takes Route A, the probability of arriving late is 10%; *supposing* Route B is taken, the chance of a late arrival rises to 40%; and *in the event that* Route C is used, the probability of being late is 25%.

(a) Can the three routes {A, B, C} form be considered as a partition of the sample? Explain.

(b) Using the Total Probability Rule, calculate the probability that the commuter arrives late on a randomly chosen workday.

(c) Given that the commuter arrived late on a particular morning, use Bayes' Theorem to find the probability that they had taken Route B.

2. A community health clinic administers a rapid diagnostic test for malaria. Every patient who visits the clinic falls into exactly one of three mutually exclusive categories: patients who are confirmed to have malaria (M_1), patients suffering from a different fever-causing illness (M_2), and patients attending a routine health check-up with no illness (M_3). Based on past clinic records, 15% of all patients have malaria, 35% have another fever-causing illness, and the remaining 50% are healthy routine attendees.

Assuming a patient has malaria, the probability of testing positive on the rapid test is 0.95. *Suppose* instead the patient has another fever-causing illness; the probability of a positive result drops to 0.12. *In the case of* a healthy routine attendee, the chance of a false positive is only 0.03.

(a) Show that {M_1, M_2, M_3} is a valid **partition** of the clinic's patient population, stating clearly the two conditions required.

(b) Using the **Total Probability Rule**, find the probability that a randomly selected patient tests positive on the rapid test.

(c) A patient has just received a positive test result. *Given this outcome*, use **Bayes' Theorem** to determine the probability that the patient actually has malaria. Briefly comment on your result.

3. A mobile phone network provider classifies its entire customer base into three non-overlapping loyalty tiers: Premium subscribers (P), Standard subscribers (S), and Pay-as-you-go users (G). Premium subscribers account for 20% of all customers, Standard subscribers make up 45%, and Pay-as-you-go users constitute the remaining 35%.

Monthly complaint records reveal the following: *if a customer is on the Premium tier*, the probability of lodging a service complaint in any given month is 0.05. *For a Standard subscriber*, this probability is 0.18. *Should the customer be a Pay-as-you-go user*, the probability of a complaint increases to 0.30.

(a) Confirm that $\{P, S, G\}$ constitutes a **partition** of the customer base. State both required conditions and verify each one explicitly.

(b) Calculate the overall probability that a randomly selected customer lodges a complaint during a given month.

(c) The customer service team has just received a complaint. *On the condition that* a complaint has been received, use Bayes' Theorem to find the probability that it was submitted by a Pay-as-you-go user. Interpret your answer in the context of the problem.

4. A bank operates an automated fraud-detection algorithm that flags transactions as suspicious. If a transaction is genuine, the probability that the algorithm incorrectly flags it as suspicious is 0.02. If a transaction is fraudulent, the probability that the algorithm flags it as suspicious is 0.90. Historical records indicate that 1% of all transactions are fraudulent.

(a) Identify the partition of a sample space this question.

(b) Using the Total Probability Rule, calculate the probability that a randomly selected transaction is flagged as suspicious.

(c) Given that a transaction has been flagged as suspicious, use Bayes' Theorem to determine the probability that the transaction is actually fraudulent. Interpret your result in the context of the bank's fraud-detection operations.